

Question	Answer	Marks
1(a)	force proportional to product of masses and inversely proportional to square of separation	B1
	idea of force between point masses	B1
1(b)(i)	velocity changes/direction of motion changes/there is an acceleration/there is a resultant force so not in equilibrium	B1
1(b)(ii)1.	gravitational force equals/is centripetal force	C1
	$GMm/R^2 = mR\omega^2$ and $\omega = 2\pi/T$ or $Gm/R^2 = mv^2/R$ and $v = 2\pi r/T$ or $GMm/R^2 = mR(2\pi/T)^2$	M1
	convincing algebra leading to $k = GM/4\pi^2$	A1
1(b)(ii)2.	correct use of R^3/T^2 for one planet (c gives 3.54×10^{21} ; e and g both give 3.56×10^{21})	C1
	$3.5(5) \times 10^{21} = (6.67 \times 10^{-11} \times M) / 4\pi^2$ $M = 2.1 \times 10^{33} \text{ kg}$	A1
	two or three values of R^3/T^2 correctly calculated and used in a valid way to find a value for M based on more than one k	B1

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Question	Answer	Marks
2(c)(i)	The value of $\frac{1}{2} \ln 2$ is 0.3466 (to 4 d.p.)	M1